

Philip-Chandan Imaging Pipeline

TCGA-BRCA longitudinal MRI: discovery through raw export and Otsu handoff

QBIHack breast-cancer-sim | Generated 2026-06-27

Philip-Chandan owns the radiomics pipeline: pull real breast MRI from TCIA, validate and stack DICOM into 3D numpy arrays, and hand raw volumes to Vinesh for PDE simulation. Resample, normalize, and segmentation are Vinesh's scope (handoff_contract.json v1.0.0, Option B).

1. Patient selection criteria

We needed real TCGA-BRCA cases for a Luminal A vs Basal-like demo with longitudinal MRI, not a single timepoint. Each candidate had to pass several gates:

Criterion	Why it matters
MRI on TCIA	Many TCGA patients have genomics but no public imaging
Longitudinal MR (>=2 study dates)	Baseline + follow-up for growth comparison
PAM50 subtype match	Luminal A and Basal-like labels align with cBioPortal
Genomics for Praneeth	ER/PR status and survival on GDC/cBioPortal
Usable contrast series	Post-contrast T1 (e.g. VIBRANT) for tumor visibility

Our first picks failed: TCGA-BH-A0BR and TCGA-A2-A04P had no MRI on TCIA. Only 19 of 139 TCGA-BRCA patients with MRI on TCIA have multiple MR studies. Patient selection was a real constraint.

Rev2 primary pair (cohort/cohort.json)

Subtype	TCGA ID	Baseline	Follow-up
Luminal A	TCGA-AR-A1AX	2002-09-12	2003-09-24 (~12 mo)
Basal-like	TCGA-AR-A1AQ	2001-11-21	2003-05-07 (~17 mo)

2. Finding patients - cohort_discovery.py

Before locking IDs in cohort.json, we built cohort/cohort_discovery.py to automate discovery and validation against public APIs:

- TCIA NBIA REST API - list MR series, group by study date, flag longitudinal cases, prefer contrast series
- cBioPortal - PAM50 subtype, ER/PR, overall survival (aligned with Praneeth's genomics fields)

CLI workflows

```
python ../cohort/cohort_discovery.py audit
python ../cohort/cohort_discovery.py find-longitudinal --subtype "Luminal A"
python ../cohort/cohort_discovery.py recommend-pair
```

Each patient gets a structured report: imaging availability, study span, PAM50 match, missing genomics fields, and a pass/fail ok flag. This tool justified the rev2 pivot and gave Praneeth the same TCGA barcodes for GDC queries.

3. Download - download_tcia.py

Once patients are locked, download_tcia.py pulls DICOM into a consistent layout:

```
data/raw/tcia/
  luminal_a/TCGA-AR-A1AX/
    2002-09-12/ # baseline
    2003-09-24/ # follow-up
  basal/TCGA-AR-A1AQ/
    2001-11-21/
```

2003-05-07/

For each study date the downloader:

- Lists MR series in collection TCGA-BRCA
- Picks the best series - prefers post-contrast (+C, VIBRANT, T1); deprioritizes calibration/localizer scans
- Downloads via idc-index (primary) with tcia-utils NBIA fallback

```
python ../download_tcia.py --all-primary --longitudinal
```

4. Validate - validate_series() in tcia_extractor.py

Download alone is not enough. Before export, validate_series(dicom_dir) checks the local series:

Check	What we catch
DICOM slices present	Empty or non-image folders
Consistent Rows/Columns	Mixed slice dimensions
Unique InstanceNumber	Duplicate or corrupted ordering
SimpleITK read succeeds	Metadata OK but ITK cannot stack
Slice count, shape, spacing_mm	Recorded in validation report for QC and sidecar JSON

extract_volume() refuses to run if validation fails. Export scripts call validate_series first and abort with explicit errors.

Visual QC

qc_slice_plot.py writes middle-slice PNGs per volume to data/qc/slice-plots-philip-chandan/, including optional overlay PNGs with a lime contour over high-intensity voxels (QC visualization only, not tumor segmentation).

Figure 1 - Spike validation slice

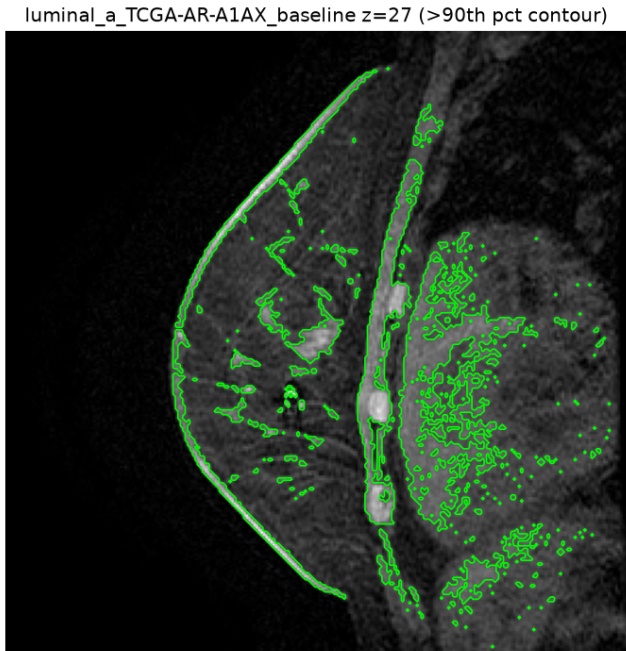


Figure 1. Raw MR slice for TCGA-AR-A1AX baseline (2002-09-12) after DICOM validation and 3D stacking. Lime contour marks voxels above the 90th percentile (QC only).

5. Export - raw volumes for Vinesh

After validation passes, we export raw 3D arrays, not PDE-ready volumes. Resample, normalize, and segmentation are Vinesh's scope per handoff_contract.json.

- Single case: export_raw_extract.py
- Batch (all primaries x timepoints): export_all_raw.py

For each patient x timepoint slug (e.g. luminal_a_TCGA-AR-A1AX_baseline):

- Resolve DICOM path from cohort.json
- Run validate_series
- extract_volume_with_spacing() - SimpleITK DICOM to float32 numpy, shape (Z, Y, X), native spacing preserved
- Write .npy, JSON sidecar, and QC plot

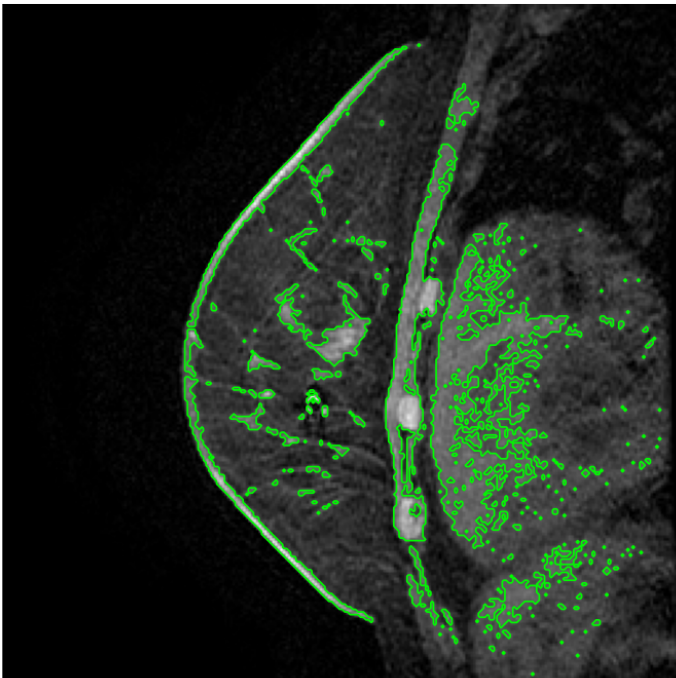
Output	Path
Raw volume	data/processed/raw-extract-philip-chandan/{slug}.npy
Metadata sidecar	.../{slug}.json (shape, spacing_mm, contract_version)
QC plot	data/qc/slice-plots-philip-chandan/{slug}_mid-z[-overlay].png
Manifest	manifest.json v1.1.0 - subtype, timepoint, slug, paths

Handoff contract - what we export vs what we do not

Philip-Chandan (raw extract)	Vinesh (PDE input)
Raw MR intensities, not normalized	Resample/crop to max 64^3, 1 mm spacing
Spacing in JSON sidecar	Normalize to [0, 1]
Axis order (Z, Y, X)	Otsu tumor segmentation

Figure 2 - Baseline subtype comparison

luminal_a_TCGA-AR-A1AX_baseline z=27 (>90th pct contour)



basal_TCGA-AR-A1AQ_baseline z=371 (>90th pct contour)

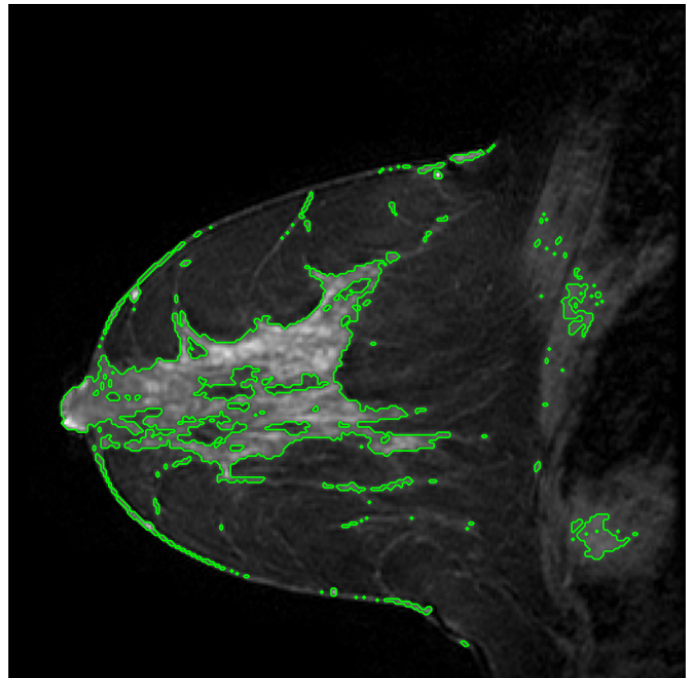
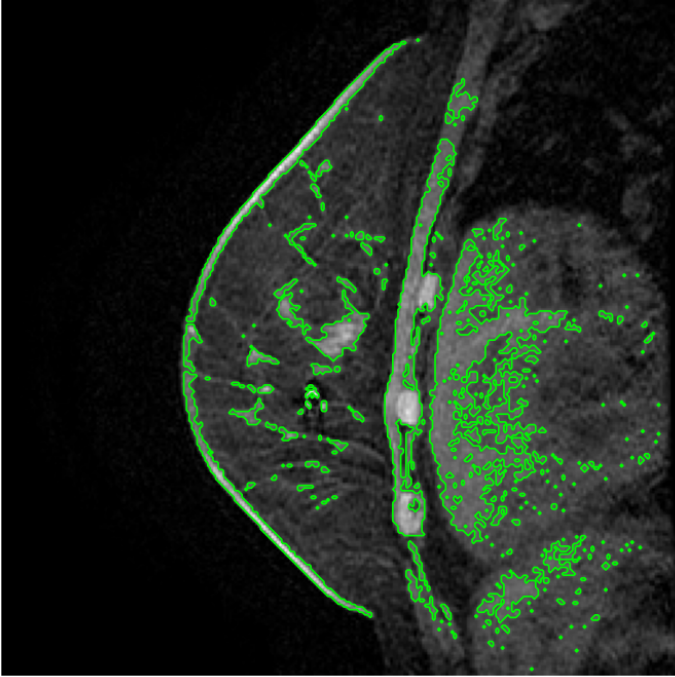


Figure 2. Baseline comparison: Luminal A (TCGA-AR-A1AX) vs Basal-like (TCGA-AR-A1AQ). Lime contour highlights enhancing tissue (90th percentile, QC only).

Figure 3 - Longitudinal Luminal A

luminal_a_TCGA-AR-A1AX_baseline z=27 (>90th pct contour)



luminal_a_TCGA-AR-A1AX_followup z=40 (>90th pct contour)

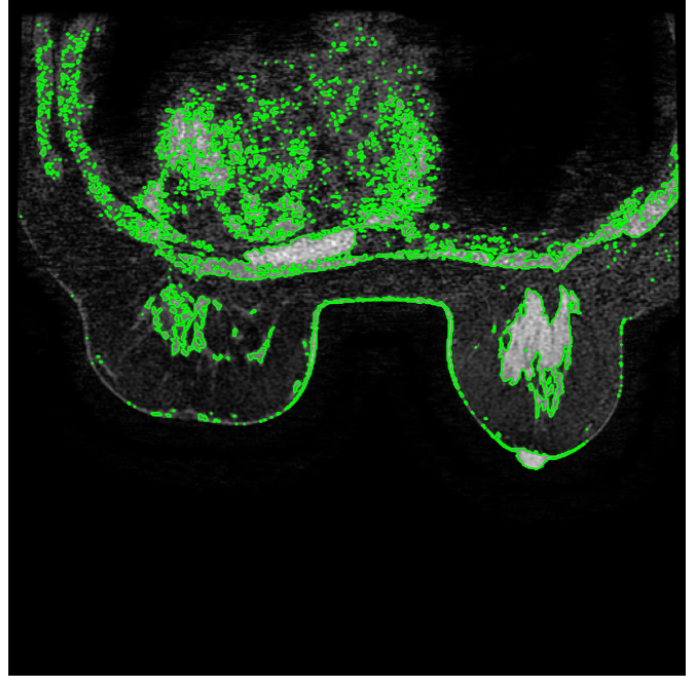


Figure 3. Longitudinal Luminal A: baseline (2002-09-12) vs follow-up (2003-09-24). Contour overlay on the slice with the most enhancing voxels per volume.

6. Otsu tumor segmentation - prepare_pde_input.py (Vinesh)

After Philip-Chandan raw export, Vinesh prepares PDE-ready volumes in vinesh/prepare_pde_input.py per handoff_contract.json v1.0.0:

- Resample raw MR to isotropic 1 mm spacing (scipy.ndimage.zoom)
- Min-max normalize intensities to [0, 1]
- Otsu threshold (skimage) on normalized voxels; keep continuous density inside tumor
- Zero background; crop/pad to max 64^3 centered on tumor center of mass

Continuous tumor values (not a binary mask) are required so the PDE logistic growth term $\rho \cdot u \cdot (1-u)$ can evolve. qc_otsu_plot.py documents the Otsu step using the same pipeline as prepare_pde_input.py.

Figure 4 - Otsu on normalized resampled volume

TCGA-AR-A1AX Otsu z=897 thresh=0.127 (mask vox=9580323)

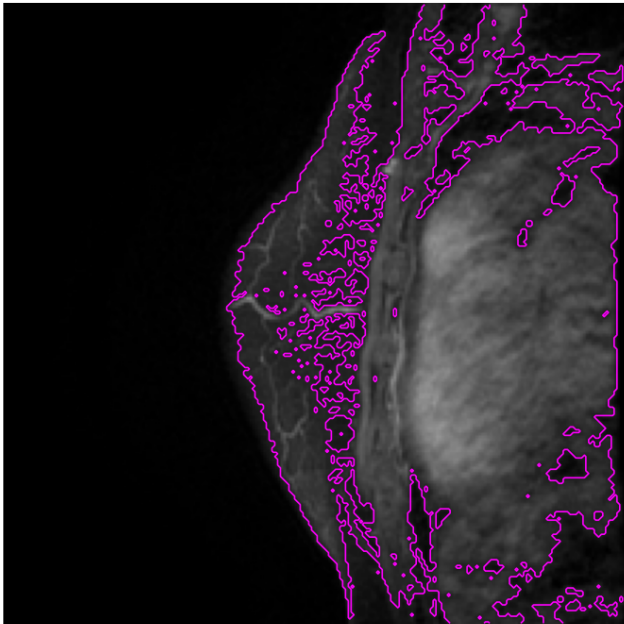


Figure 4. TCGA-AR-A1AX baseline after resample + normalize. Magenta contour is the Otsu tumor mask (voxels above threshold); background will be zeroed before crop.

Figure 5 - PDE input after Otsu and crop

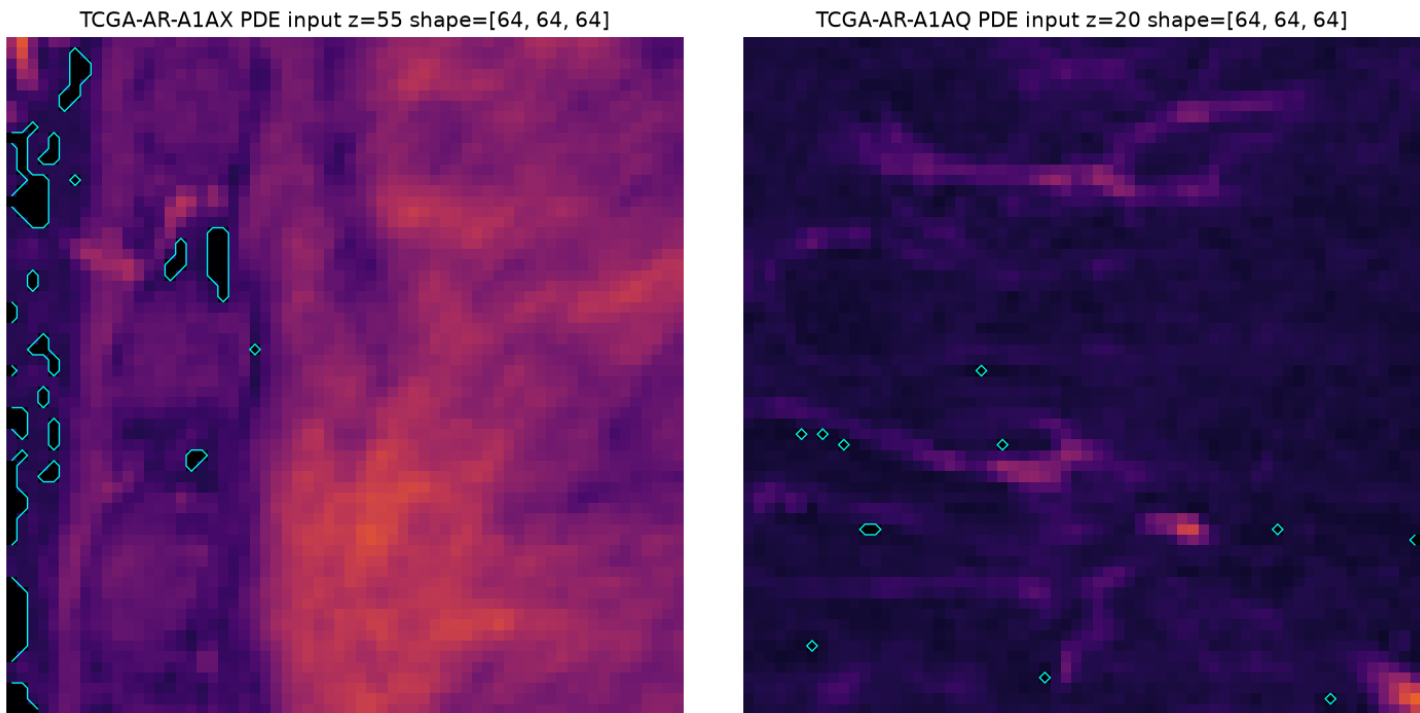


Figure 5. Baseline PDE inputs (64^3 , 1 mm) for Luminal A vs Basal-like. Inferno colormap shows continuous initial tumor burden; cyan contour marks tumor voxels > 0 .

7. Segmentation validation vs TCIA radiologist masks

TCGA-Breast-Radiogenomics publishes expert lesion segmentations (*.les) for 91 patients. We downloaded the public ZIP (~103 KB), parsed cuboid masks, and compared them to our Otsu + largest-component heuristic on the matching DCE series (S2 = VIBRANT for both rev2 baselines). Follow-up timepoints have no .les annotations.

Slug	Dice	Area (Otsu/.les)	Expert vol	Otsu vol
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luminal a TCGA-AR-A1AX baseline	0.000	2649.9x	2,927	7,756,222
basal TCGA-AR-A1AQ baseline	0.000	619.4x	6,075	3,762,799

Dice is near zero on both primaries: Otsu selects a much larger bright region than the radiologist ROI. Area fraction (Otsu voxels / .les voxels) is ~2650x for Luminal A and ~620x for Basal-like. Treat Otsu masks as demo heuristics, not clinician-equivalent segmentations.

Figure 6 - Radiologist .les vs Otsu (baseline)

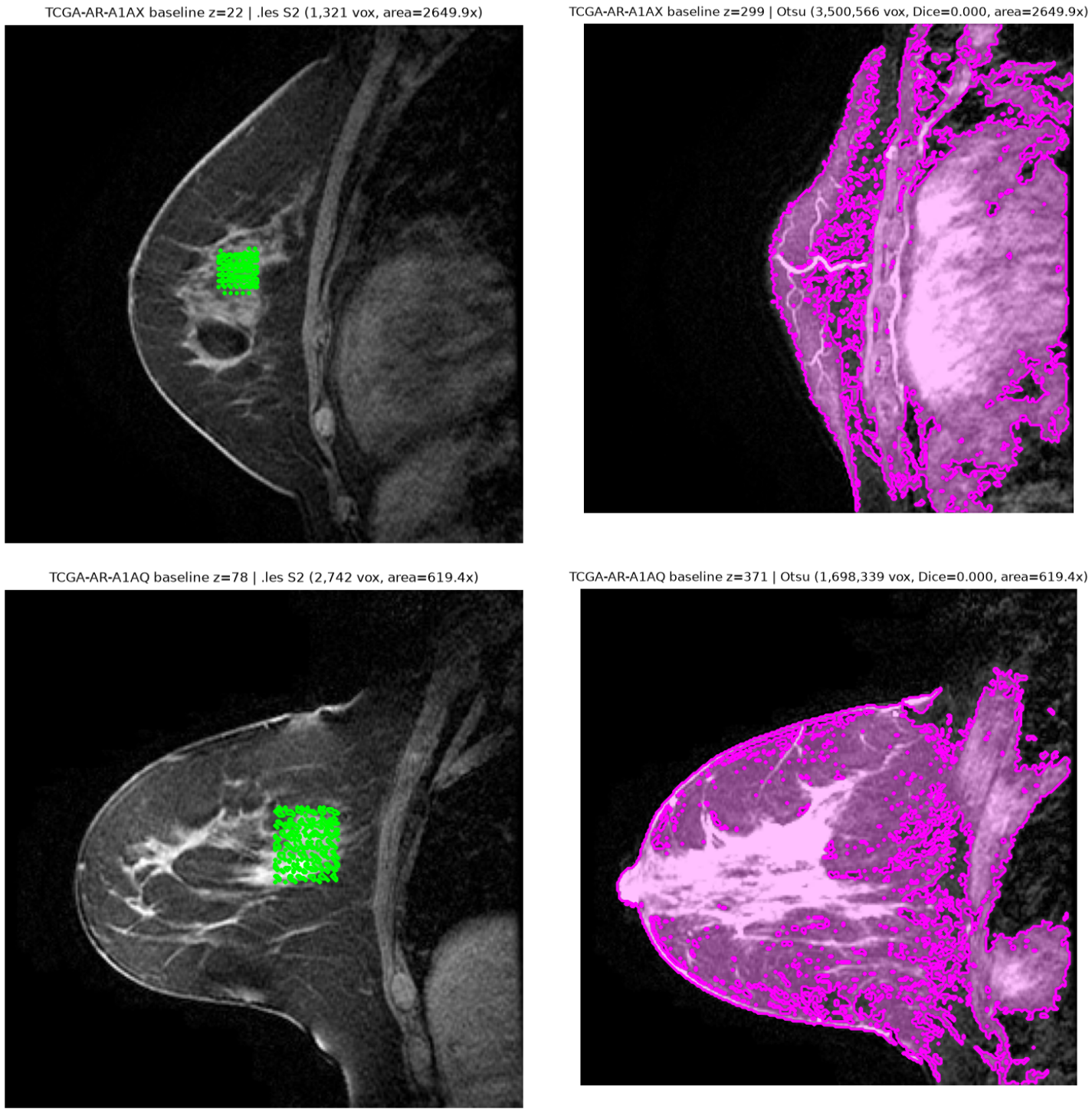


Figure 6. Baseline validation (2x2): top row = Luminal A .les then Otsu; bottom row = Basal-like .les then Otsu. Green = TCIA .les; magenta = Otsu. Each panel uses its own best z slice (masks do not overlap in slice space). Titles show voxel counts and area=Otsu/.les.

8. End-to-end summary

We defined strict patient criteria (longitudinal MRI on TCIA, PAM50-aligned subtypes, genomics for the team), built `cohort_discovery.py` to search and audit candidates against TCIA and cBioPortal, locked rev2 primaries after rev1 failed imaging availability, downloaded four DICOM timepoints with contrast-aware series selection, validated each series structurally before extraction and visually via slice QC, exported four raw `.npy` volumes plus JSON sidecars and a manifest, documented Vinesh Otsu segmentation into 64^3 PDE inputs for downstream simulation, and benchmarked Otsu against TCIA radiologist `.les` masks on baseline VIBRANT (Dice near zero - heuristic only).

Deliverables

Stage	Deliverable	Count
Discovery	cohort.json rev2 + COHORT.md	2 primaries + backups
Download	DICOM under data/raw/tcia/	4 study folders
Validate	validate_series pass + QC PNGs	4 volumes
Export	Raw <code>.npy</code> + <code>.json</code> + manifest.json	4 slugs
PDE prep	Otsu segmentation QC + pde-input-vinesh/	4 slugs
Seg. validation	<code>.les</code> comparison + validation_metrics.csv	2 baselines

Exported volume shapes (rev2)

Slug	Shape (Z,Y,X)	Spacing (mm)
luminal_a_TCGA-AR-A1AX_baseline	[352, 256, 256]	[3.0, 0.8594, 0.8594]
basal_TCGA-AR-A1AQ_baseline	[464, 256, 256]	[3.0, 0.859375, 0.859375]
luminal_a_TCGA-AR-A1AX_followup	[552, 512, 512]	[2.2, 0.5273, 0.5273]
basal_TCGA-AR-A1AQ_followup	[448, 256, 256]	[3.0, 0.9375, 0.9375]